Ruben Perez

+1 408-430-4740 • perezben37@gmail.com • linkedin.com/in/rubaduben

Website:

Summary

Enthusiastic and results-driven Computer Engineer with a strong robotics and software engineering background. I am seeking to leverage my expertise in autonomous systems and software development to contribute to innovative projects in the field of robotics.

Education

Bachelor of Science | ***Computer Engineering with a Concentration in Robotics*** — September 2017 – June 2021
University of California, Santa Cruz — Santa Cruz, CA

Work Experience

Zipline — South San Francisco, CA
Senior Hardware Test Automation Engineer — October 2024 – Present

- Developed hardware test systems for autonomous drones and dock systems by automating flight profiles, sensor checks, and safety protocols using custom Python and gRPC interfaces.
- Engineered a modular finite state machine framework for embedded debugging and streamlined state management.
- Containerized test environments with reproducible PEX binaries and automated Ansible deployments.
- Integrated custom instrumentation for real-time monitoring and control of diverse hardware (e.g., power supplies, DAQs, pneumatic/hydraulic controllers, IMUs).

Advanced Farm — Davis, CA
Software Engineering Manager of Testing — October 2022 – October 2024

- Architected distributed compute systems and implemented real-time Python microservices on Linux for autonomous fruit harvesting platforms, enhancing scalability and reliability.
- Developed an automated apple harvester drive system with custom PID controllers, asynchronous state machines, and CAN/CANOpen protocols to improve navigation and efficiency.
- Migrated the compute and vision stack to a unified Nvidia Jetson Orin AGX platform, Improving performance and simplifying maintenance.
- Led a cross-functional team to optimize subsystem performance and ensure robust field operations.

└ ***Software Test Engineer*** — November 2021 – October 2022

- Commissioned and validated a fleet of robotic strawberry harvesters, ensuring integration of firmware and components.
- Optimized and took ownership of a critical robotic sub-system within three months, Effectively reducing position following error by a factor of 5 and greatly improved driveability in rough terrain. ***Python***, ***Twisted***, and ***ZeroMQ***.
- Developed Python command-line tools to decode CAN traffic into CANopen messages and visualize sensor data.

└ ***Software Support Engineer*** — March 2021 – November 2021

- Monitored a fleet of strawberry harvesters in the field, providing software support and conducting maintenance to optimize performance and minimize downtime.

Skills

Languages: English (Fluent), Spanish (Native)
Programming: Python (NumPy, SciPy, Matplotlib, Twisted, ZMQ, Asyncio, gRPC, CAN/CANopen), C/C++, MATLAB, Bash, Git, HTTP
Software: ROS, Linux, Gazebo, Docker, Ansible, Debian Packaging, Graphviz, stm32 CubeIDE, Fusion 360, Github
Instrumentation & Embedded: NI cDAQ, LabJack, custom hardware interfacing, real-time control, distributed microservices